

* Decreasing the average length of the code word. Arithmetic Coding: $\frac{2^n \rightarrow 2^{n+1}}{1m \text{ iterations}}$

Q. Consider 3 letters, $S = \{1, 2, 3\}$ $\begin{cases} * 2^n = 2^{n-1} + (u^{n-1} - 2^{n-1}) F_X(x_{n-1}) \\ * u^n = 2^{n-1} + (u^{n-1} - 2^{n-1}) F_X(x_n) \end{cases}$
 respective $P(1) = 0.5, P(2) = 0.3, P(3) = 0.2$
 Encode: $1332 \rightarrow a_1, a_3, a_3, a_2$
 Where: $|n|$ = length of the sequence going to decode.
 $|n|$ = sequence is to encode.
 $n = 0, 1, 2, 3, \dots$

Continuous go on finding from 2^0 to 2^4 as there four nos to decode. (four letters to decode)

$\rightarrow 2^0 = 0$ | $u^0 = 1$ Default value.

lower limit | upper limit.

$F(x) = \text{Cumulative sum of the probabilities}$
 $\rightarrow F(0) = 0$ $F(1) = P(1)$ $F(2) = \text{sum of } P(1) + P(2)$ $F(3) = P(1) + P(2) + P(3)$
 $= 0$ $= 0.5$ $= 0.5 + 0.3 = 0.8$ $= 0.5 + 0.3 + 0.2 = 1.0$

using Formula: $n=1$ $x_{n-1} \rightarrow$ as $[1, 3, 3, 2]$

$\rightarrow 2^1 = 2^0 + (u^0 - 2^0) \cdot F_X(1-1)$

$n=1$ $= 0 + 1(0) = 0$

$\rightarrow u^1 = 2^0 + (u^0 - 2^0) \cdot F_X(1)$

$= 0 + (1) \cdot 0.5 = 0.5$

$\rightarrow 2^2 = 2^1 + (u^1 - 2^1) \cdot F_X(3-1)$

$n=2$ $= 0 + (0.5)(0.8) = 0.4$

$u^2 = 2^1 + (u^1 - 2^1) \cdot F_X(3) = 0 + (0.5)(1) = 0.5$
 $x_{n-1} = 3$ $[1, 3, 3, 2]$

$\Rightarrow 2^3 = 2^2 + (u^2 - 2^2) \cdot F_X(3-2) = 0.4 + (0.5 - 0.4)(0.8) = 0.48$

$n=3$ $u^3 = 2^2 + (u^2 - 2^2) \cdot F_X(3) = 0.4 + (0.5 - 0.4)(1) = 0.5$
 $x_{n-1} = 2$ $[1, 3, 3, 2]$

$\Rightarrow 2^4 = 2^3 + (u^3 - 2^3) \cdot F_X(1) = 0.48 + (0.5 - 0.48)(0.5) = 0.49$

$n=4$ $u^4 = 2^3 + (u^3 - 2^3) \cdot F_X(2) = 0.48 + (0.5 - 0.48)(0.8) = 0.496$

1. Last step: - Encode 1332, Unencoded value: -

find the average of 2^4 & u^4

average of $\frac{2^4 + u^4}{2} = \frac{0.49 + 0.496}{2} = 0.493$ Unencoded tag.
 using arithmetic coding.
 $\rightarrow$ Binary.